Learning Instance-Level Knowledge With Image-Level Supervision for Open-Vocabulary Object Detection in Remote Sensing Images

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***Abstract*—Open-vocabulary object detection (OVOD) has shown significant potential in real-world remote sensing applications, thanks to its adaptability to open category spaces. Existing OVOD methods mainly rely on pretrained vision-language models (VLMs) to recognize unknown categories. However, VLMs are usually trained on image–text pairs. Whilst possessing strong global semantic modeling capabilities, they struggle to accurately capture the relationship between object instances and their precise locations. To address this problem, this article presents LILK, which learns instance-level knowledge from image-level data through weak supervision. Specifically, an image-level weakly supervised knowledge injection (IWKI) module is first created, which represents instance-level semantic information by introducing global classification queries, while aligning image-level supervision with instance-level supervision through a distinct query promotion (DQP) strategy. Second, a quality-aware pseudo-label rectification (QPLR) module is developed, which filters candidate boxes using dual thresholds based on detector confidence and semantic consistency, while incorporating a SAM2-based relocation mechanism to enhance spatial localization accuracy of pseudo labels. Finally, an image-guided query enhancement (IGQE) module is introduced to guide and reinforce detection queries using instance-level image features, providing the detector with stable and modality-consistent visual References. Extensive experimental results on three commonly used remote sensing object detection datasets, DIOR, DOTA, and NWPU VHR-10, demonstrate the efficacy of LILK in performing OVOD tasks.**

***Index Terms*—Object detection, open vocabulary, remote sensing imagery.**

I. Introduction

With the rapid development of remote sensing imaging technology and Earth observation capabilities in addressing real-world problems such as land surveys, urban planning, traffic monitoring, and disaster assessment, a massive number of high-resolution remote sensing images are continuously being acquired. As one of the core tasks in remote sensing image understanding, object detection aims to automatically locate and identify objects of interest from complex backgrounds, offering significant value in practical applications. However, most existing mainstream remote sensing object detection methods [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14] are based on the closed-set assumption, meaning that the set of object categories in the training phase remains the same as in the testing phase. Although this setting may lead to good performance in controlled experimental environments, it often fails to meet the needs of practical remote sensing applications: new types of targets emerge constantly, and category definitions change frequently according to application requirements, while recollecting and annotating data as well as retraining models are both costly and inefficient. To overcome such limitations of closed-set assumptions, open-vocabulary object detection (OVOD) [15], [16], [17], [18], [19], [20] has gradually become an important research direction in computer vision. Its goal is to enable detection models to recognize categories that did not appear during training, thereby achieving effective generalization from known categories to unknown ones.

The current mainstream OVOD methods work mainly by relying on the presumed cross-modal semantic representation capability of pretrained vision-language models (VLMs) [21], [22], [23], [24] in supporting the recognition of unknown categories. Typical methods include PromptDet [16], which adopts a two-stage framework: class-agnostic object localization is performed first, followed by classification of candidate regions using a VLM. BARON [17] incorporates VLMs for object recognition while aligning a bag of regions with text features to enhance the detector's semantic understanding in complex scenes. Sambor [18] integrates the generalization capabilities of CLIP [21] and SAM [25], exploiting these two component methods for candidate region classification and class-agnostic object localization, respectively. For remote sensing scenarios, InstructSAM [19] applies SAM2 [26] to generate class-agnostic candidate boxes and combines with a domain-specific VLM (purposely devised for remote sensing images) to classify candidate regions, thereby achieving OVOD.

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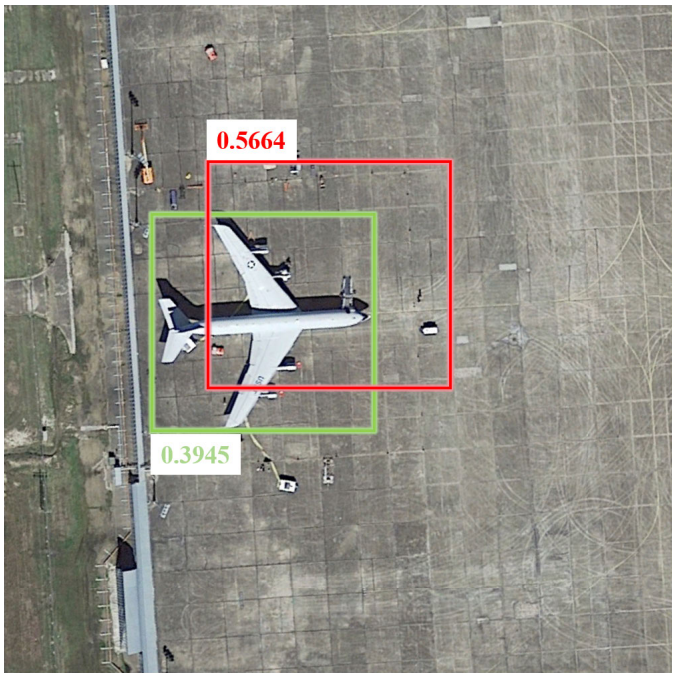


Fig. 1. Example of semantic matching results between different candidate boxes and "airplane" in Git-RSCLIP [24]. Although the green box is more accurate in spatial localization, its semantic matching score is lower than that of the red box with poorer localization quality.

Although VLM-based methods have made remarkable progress in OVOD, their inherent training paradigm restricts the performance of instance-level localization. VLMs trained on large-scale image–text pairs demonstrate strong capability in modeling global image semantics, but they lack fine-grained instance-level alignment. As a result, VLMs often fail to precisely associate specific objects with their regions, leading to suboptimal localization performance in OVOD, as illustrated in Fig. 1. As an earlier approach to address this problem, RegionCLIP [27] attempts to enhance regional perception by fine-tuning CLIP using region–text pair data. However, this strategy still suffers from two main limitations: 1) region–text pairs typically rely on pretrained CLIP to match candidate regions and text, making it difficult to ensure that the selected regions have accurate spatial localization; and 2) fine-tuning CLIP risks damaging its original general-purpose cross-modal representation capability, thereby weakening its generalization performance in open-vocabulary tasks.

This article proposes a novel OVOD method to learn instance-level knowledge from image-level annotated data in a weakly supervised manner. Abbreviated to LILK, it consists of the following three key modules.

1) Image-level weakly supervised knowledge injection (IWKI), which introduces global classification queries combined with multiobjective classification loss to model instance-level semantic information without relying on instance-level annotations, and designs a distinct query promotion (DQP) strategy to effectively align image-level supervision with instance-level queries.

2) Quality-aware pseudo-label rectification (QPLR), which performs dual-threshold joint filtering using detector confidence and semantic consistency as per Git-RSCLIP [24] and proposes a dynamic dual-threshold scheduling strategy adapting to different pseudo-label quality requirements across training stages, upon which to introduce a relocation mechanism built on SAM2 [26], to reinforce the spatial localization accuracy of pseudo-labels via multipoint prompt segmentation.

3) Image-guided query enhancement (IGQE), which uses pre-extracted instance-level image features to guide and enhance detector queries, providing the overall model with stable and modality-consistent visual references to improve object instance localization and recognition.

To validate the effectiveness of the proposed method, extensive experiments are conducted on three popularly adopted, representative remote sensing object detection datasets (DIOR, DOTA, and NWPU VHR-10). Experimental results show that LILK achieves state-of-the-art performance across all these datasets, demonstrating the effectiveness and potential of learning instance-level knowledge from image-level supervision in remote sensing OVOD.

The main contributions of this article can be summarized as follows.

1) A new remote sensing OVOD framework, LILK, is proposed. Unlike existing methods that mainly rely on VLMs for instance-level recognition and localization, LILK learns instance-level knowledge from image-level annotated data through weak supervision.

2) An IWKI module is designed. It achieves effective transfer from image-level supervision to object-level semantics without introducing instance-level annotations, through global classification queries and a DQP strategy.

3) A QPLR module is developed, which combines detector confidence with visual-language semantic consistency. This is supported by the proposal of a dynamic dual-threshold scheduling strategy and the introduction of a SAM2-based relocation mechanism, generating high-quality pseudo-labels that are both spatially precise and semantically reliable.

4) An IGQE module is created to reinforce detection stability and localization accuracy. By leveraging pre-extracted instance-level image features to guide and enhance detector queries, this strategy provides a stable and modality-consistent visual reference throughout the object detection process.

The remainder of this article is arranged as follows. Section II presents an overview of related work. Section III details the proposed methodology and associated techniques. Section IV shows the results of experimental investigations and discusses experimental results. Finally, Section V concludes this article and points out directions for further research.

II. Related Work

A. Pretrained VLMs

In recent years, the rapid development of pretrained VLMs [21], [22], [23], [24], [28], [29], [30], [31], [32], [33] has greatly promoted research progress in zero-shot learning and open-vocabulary visual tasks. By performing contrastive learning on large-scale image–text pair data, a VLM can construct a unified cross-modal representation space, effectively aligning images and text at the semantic level. This enables image understanding and classification based on language prompts,

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providing strong prior support for zero-shot learning and open-vocabulary tasks.

CLIP [21] is a pioneering work in the approaches to building VLMs. It adopts separate image and text encoders, mapping the two modalities into a shared semantic space, and aligns them cross-modally through contrastive learning loss. CLIP has demonstrated strong generalization capabilities across various downstream tasks, sparking extensive research on VLMs. Building on the CLIP framework, ALIGN [28] further expands the scale of training data; despite introducing more noisy data, it can achieve significant performance improvements. SigLIP [22] replaces CLIP's softmax-based contrastive learning loss with a more flexible pairwise sigmoid loss function, effectively reducing dependence on large batch sizes and strengthening training stability and scalability. BLIP [29] also covers vision-language understanding and generation tasks on the basis of the contrastive learning paradigm. By optimizing the image–text contrastive loss, the image–text matching loss, and the language modeling loss, it supports multiple tasks such as image caption generation and visual question answering simultaneously. Furthermore, methods such as Flamingo [30] and LLaVA [31] combine pretrained visual encoders with large-scale language models, enabling a language model to process multimodal inputs, significantly expanding the application scope of visual-language models.

With the significant success of VLMs in handling natural images, this paradigm has been introduced into the realm of remote sensing imagery to address challenges such as high annotation costs and complex category distributions. RemoteCLIP [32] is one of the earlier VLMs designed for remote sensing scenarios. Based on the CLIP framework, it converts remote sensing object detection and semantic segmentation datasets into image description form, alleviating the scarcity of remote sensing image–text pairs. SkyCLIP [33], targeting the diversity of scene types and complex semantic distributions in remote sensing, improves the construction of image–text pairs by exploiting richer scene-level textual descriptions, thereby enhancing the model's ability to represent remote sensing scene semantics. GeoRSCLIP [23], from the perspective of geographic semantic consistency, incorporates prior information like geographic location and scene attributes into the construction of image–text pairs to improve the alignment quality between visual representations and remote sensing semantics. Git-RSCLIP [24] further advances the scaling of remote sensing vision-language pretraining datasets, substantially increasing the number of pretraining images to the tens of millions. It enables the model to exhibit stronger semantic representation capabilities and greater generalization potential in tasks such as remote sensing image understanding and cross-modal retrieval.

Although VLMs in the field of remote sensing have made significant progress, their overall capabilities remain restricted as compared to large-scale VLMs in the natural image domain due to limitations in the size of available pretraining data and model capacity. This situation calls for further research on how to more effectively leverage vision-language priors in remote sensing scenarios and integrate them with instance-level visual tasks.

B. Open-Vocabulary Object Detection

With the breakthrough progress of VLMs in cross-modal semantic alignment, investigations of object detection have been carried out into paradigms beyond traditional closed-set classifications. In particular, OVOD [15], [16], [17], [18], [19] has gradually emerged. It aims to enable detection models to identify novel object categories that did not appear during the training phase, significantly enhancing the generalization capability of object detection systems in dealing with real-world problems.

Early OVOD methods mostly adopt a two-stage paradigm, with OVR-CNN [15] being one of the representative approaches. It first learns a VLM using image-caption pairs and then applies this VLM to an object detection method for OVOD. Owing to its clear structure and ease of integration with VLMs, such a two-stage paradigm has become the foundational framework for many subsequent OVOD methods. PromptDet [16] uses a class-agnostic proposal generator to produce candidate regions first, followed by leveraging a VLM (e.g., CLIP) to perform prompt-based classification of the candidate regions. RegionCLIP [27] illustrates that directly utilizing CLIP for pretraining on entire images may fail to meet the fine-grained matching requirements between region and text concepts. Therefore, it proposes a region-level extension to CLIP by introducing region–text pairs for fine-tuning, enabling the model to learn the alignment between region-level visual representations and text semantics. The method applies this technique to the classification of candidate boxes generated by the RPN [34]. CORA [35] explores OVOD of the DETR-style [36], alleviating the distribution gap between global features and regional features through region prompts. It also introduces a class-aware matching mechanism to enhance object localization capabilities, thereby improving the adaptability of CLIP in end-to-end detection frameworks. Sambor leverages the strong generalization ability of SAM [25] in segmentation to generate class-agnostic candidate regions and then exploits CLIP's vision-language alignment capability to perform open-vocabulary classification.

In addition to directly using VLMs for region classification, techniques exist that attempt to mine object-level supervision signals from image–text pair data. PB-OVD [37] generates pseudo labels from image–text pairs to assist object detection training. This method utilizes selective search [38] to generate candidate regions. It relies on the cross-modal alignment capability of VLMs to filter and annotate the candidate boxes, enabling the detector to learn object-level knowledge from image-level data. VL-PLM [39] improves foreground object localization accuracy by introducing an iterative object positioning mechanism in the detection head to refine candidate object locations. It leverages a VLM to perform semantic classification on precisely localized object instances. WSOVOD [40], from the perspective of weakly supervised object detection, combines RPN [34] and SAM [25] to generate candidate regions. It corrects and optimizes the detector through data-aware features, while employing a VLM to complete category prediction for the candidate regions.

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Another category of methods explicitly incorporates linguistic information into object detectors to align instance features with category language features, achieving OVOD. Representative approaches of this category include GLIP [41], Grounding DINO [42], and YOLO-World [43]. These methods typically use language embeddings as conditional information and jointly represent them with image features during detection, enabling the detection model to directly perceive linguistic semantics within an end-to-end framework.

With the rapid development of OVOD in the field of natural images, other related concepts have also been applied to remote sensing image scenarios. For instance, LLaMA-UniDetector [44] follows the two-stage paradigm of PromptDet, introducing large-scale VLMs such as LLaMA [45] into the second stage to perform semantic discrimination on candidate regions. InstructSAM [19] presents a training-free OVOD method for remote sensing images: It first uses large models like Qwen [46] to infer the potential categories and their quantities in the image, followed by employing SAM2 [26] to generate category-agnostic candidate regions, and finally, it applies GeoRSCLIP [23] to classify the candidate boxes. CastDet [47] explores learning novel category knowledge through a pseudo-labeling strategy on unlabeled remote sensing images. To ensure the quality of pseudo-labels, it employs a novel dual-teacher architecture: a localization teacher generates candidate bounding boxes, while an external teacher leverages RemoteCLIP [32] to semantically classify and refine these generated boxes.

LILK differs from prior pseudo-label-based open-vocabulary detectors such as WSOVOD and VL-PLM in both detector architecture and learning paradigm. WSOVOD and VL-PLM are built on Faster R-CNN-style frameworks, where class-agnostic proposals are generated first and then categorized with the aid of VLMs. In contrast, LILK is developed on DDQ-DETR, addressing a different challenge: enabling a DETR-style detector to exploit image-level supervision for novel categories. Accordingly, LILK introduces a DDQ-DETR-oriented weakly supervised learning pipeline, where IWKI learns image-level novel-category awareness and transfers such awareness to instance-level distinct queries, with QPLR further refining pseudo labels through semantic correction and localization optimization. Therefore, LILK is not a direct extension of proposal-based pseudo-label assignment methods, but a weakly supervised adaptation framework tailored to DETR-style detection.

LILK also differs from CORA. Although both are associated with DETR-style open-vocabulary detection, CORA mainly improves open-vocabulary generalization by aligning local region features with CLIP-derived semantics. In contrast, LILK focuses on learning novel-category knowledge from image-level annotations and progressively transforming it into instance-level detection capability through weakly supervised knowledge transfer and pseudo-label refinement. Thus, while CORA primarily follows a feature-alignment paradigm, LILK is centered on weakly supervised knowledge transfer and refinement for DDQ-DETR.

III. Methodology

A. Preliminary

Within the open-vocabulary detection paradigm, an object detector is typically trained on data containing base classes C_{base}, while at test time it is expected to be able to detect all objects from both base classes C_{base} and novel classes C_{novel}, with the intersection between the base and novel classes being empty, that is, $C_{base} \cap C_{novel} = \emptyset$.

In this work, DDQ-DETR [48] is used as the baseline and systematically extended for open-vocabulary detection scenarios, as shown in Fig. 2. Specifically, the image encoder of SigLIP2 [49] is adopted as the visual backbone network for DDQ-DETR, while utilizing the text encoder of SigLIP2 to extract text features that correspond to categories. Based on these text features, a classifier is constructed to replace the original closed-set classification head of DDQ-DETR, thereby endowing the model with open-vocabulary recognition capability. On this basis, three mutually synergistic key modules are further proposed to enhance the overall capability of the resulting model for open-vocabulary detection. The key properties of these three modules are outlined here, with their implementation details to be specified afterward as follows.

1) *IWKI Module:* This module introduces global classification queries into the decoder and combines them with the DQP strategy, guiding the detector to effectively align image-level category supervision with instance-level detection queries. It enables the integrated model to learn discriminative object-level representations from image-level data, even in the absence of bounding box annotations.

2) *QPLR Module:* This module utilizes the detector to perform inference on image-level data, generating candidate object-level pseudo-labels. The pseudo-labels are filtered through a dual-threshold joint filtering strategy that combines detector confidence and Git-RSCLIP [24] semantic scores. A dynamic dual-threshold scheduling mechanism is designed for adaptation of the model's evolving capabilities during different training phases. A pseudo-box relocation mechanism based on SAM2 [26] is further introduced to enhance the spatial accuracy and reliability of pseudo-labels.

3) *IGQE Module:* This module explicitly guides decoder queries with image features from the perspective of visual similarity. By introducing same-modality visual priors at the query level, it effectively narrows the gap between the guiding information and the detection feature space, thereby further strengthening the model's ability to perceive and distinguish target instances in open-vocabulary scenarios.

Through the synergistic effect of the above three modules, the resulting object detector can progressively bootstrap high-quality object-level knowledge from image-level supervision and, guided by image feature priors, achieve robust detection of novel categories and targets.

B. Image-Level Weakly Supervised Knowledge Injection

Existing OVOD methods typically rely on external prior knowledge obtained from large-scale VLMs (e.g., CLIP [21]). However, such prior knowledge is mainly learned through

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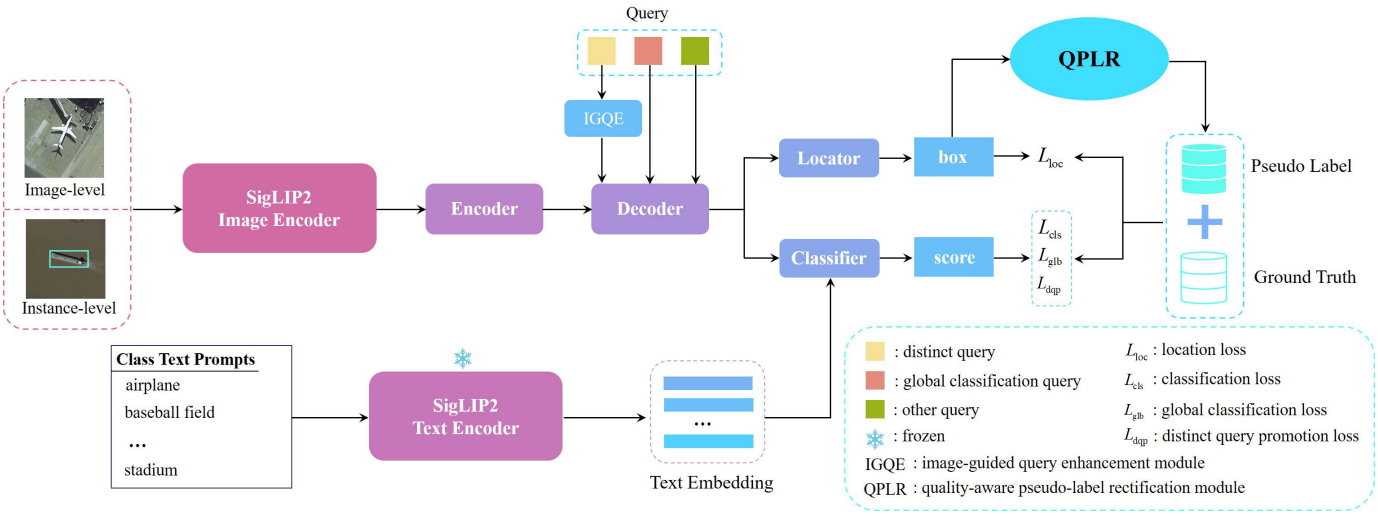


Fig. 2. Overview of LILK. It consists of three main innovative parts: 1) IWKI, learning instance-level knowledge by global classification query, global classification loss, and DQP loss; 2) QPLR, improving the quality of pseudo-labels through dual-threshold joint filtering and SAM2-based relocation; and 3) IGQE, using instance-level image features to guide and enhance detector queries.

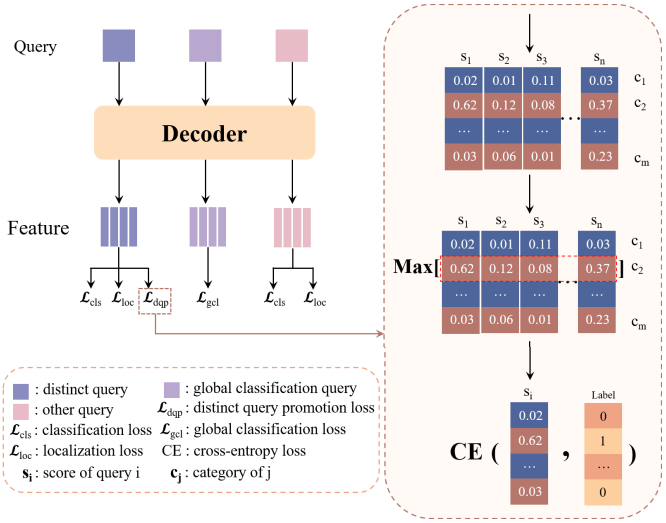


Fig. 3. Illustration of IWKI.

image-level alignment (e.g., image–text contrastive learning [21], [50]) or cross-task supervision (e.g., VQA [51], image captioning [52]), which differ significantly from object-level detection tasks in terms of supervision granularity and optimization objectives. When this knowledge is directly transferred to performing object detection, it often introduces issues such as semantic bias and imprecise localization [27]. In the context of remote sensing imagery, these problems are normally amplified due to complex backgrounds, large-scale variations, and densely distributed objects.

To alleviate this misalignment issue, an IWKI module is proposed to learn transferable detection-related semantic knowledge from image-level annotated data, as shown in Fig. 3. This is based on the DDQ-DETR [48] detection framework, jointly utilizing object-level detection data and image-level classification data for training.

In the decoder of DDQ-DETR, the original design includes different types of queries, including: distinct queries, used for performing one-to-one matching between predicted boxes and ground truth, while undertaking the primary object detection task; and dense queries and denoising queries, serving as auxiliary queries to stabilize and accelerate the training process.

On this basis, a global classification query is added to represent instance-level semantic information.

Particularly, the global classification query q_{glb} is defined as a learnable embedding vector

$$q_{glb} = Embedding\,(d)\,, q_{glb} \in R^d \tag{1}$$

where $Embedding(\cdot)$ represents a randomly initialized learnable embedding, and d is the dimension of the query embedding. The global classification query is concatenated with the original distinct, dense, and denoising queries. Together, they are fed into the decoder for feature decoding, thereby obtaining the corresponding global classification feature representation

$$q_{all} = \left[q_{dist}, q_{dens}, q_{deno}, q_{glb}\right] \tag{2}$$

$$F_{all} = \left[F_{dist}, F_{dens}, F_{deno}, F_{glb}\right] \tag{3}$$

where $F_x = Decoder(q_x), x \in dis, dens, deno, glb$, with $dis, dens, deno, glb$ being the abbreviations of distinct, dense, denoising, and global classification query, respectively. The decoded global classification features are fed into a classifier based on text feature matching, which outputs the prediction logits l_{glb}

$$l_{glb} = Classifier\left(F_{glb}\right). \tag{4}$$

Since remote sensing images usually contain multiple categories simultaneously, image-level supervision is herein treated as a multilabel classification problem. It adopts the sigmoid activation function together with the binary cross-entropy (BCE) loss for optimization

$$L_{glb} = BCE\left(sigmiod\left(l_{glb}\right), y\right) \tag{5}$$

where $y \in \{0,1\}^C$ is the multihot encoding representing the combination of categories present in the image, and C is

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the total number of categories. Through global classification queries, the model can directly learn category-related visual representations from image-level annotation data, thereby enhancing the detector's perception of the overall semantic meaning of the image concerned.

Despite image-level classification supervision, the lack of instance-level annotations in image-level data often makes it difficult for detection queries to obtain sufficiently strong positive gradient signals during training. This can result in low prediction confidence of the model for instances in image-level data. To address this, the DQP strategy is further proposed to achieve alignment between image-level supervision and object-level detection queries.

Given an image, the set of categories C_{img} it contains is known. For any given category $c \in C_{img}$, among all distinct queries' prediction results, suppose that the query with the highest predicted confidence for that category is selected, such that

$$q^* = \arg\max_{q_i} p(q_i, c) \tag{6}$$

where $p(q_i, c)$ represents the prediction score of the ith distinct query for category c. Then, this query can be treated as a pseudo-positive sample for that category, applying supervision only to its classification branch. Its category label is therefore, set as the one-hot encoding of the corresponding category with the classification loss calculated as

$$L_{dpq} = CE(p(q^*), y_c) \tag{7}$$

where $CE(\cdot)$ denotes the cross-entropy loss. Through this strategy, image-level category information can be explicitly transmitted to object-level detection queries, promoting the response of detection queries to real targets within the image without instance annotations.

C. Quality-Aware Pseudo-Label Rectification

To further exploit the object-level information within image-level annotated data, an inference is performed on the image-level data after each training epoch, generating the corresponding object-level pseudo-labels. Specifically, the detector trained in the current epoch is used to infer all image-level data, producing a set of bounding boxes and category confidence scores. Since the quality of pseudo-labels affects the stability and performance of subsequent training, the present detection results are not used directly. Instead, a QPLR mechanism is developed to ensure the semantic and localization reliability of the generated pseudo-labels, as shown in Fig. 4.

Relying solely on detector confidence may, however, introduce noisy pseudo-labels, especially in the early stages of model training. To address this issue, Git-RSCLIP [24] is exploited as an external semantic consistency evaluation module to independently score the semantic quality of bounding boxes. For each bounding box, two types of scoring information are obtained: the prediction confidence s_{det} from the object detector and the semantic matching score s_{clip} from Git-RSCLIP. By utilizing these two types of information, a

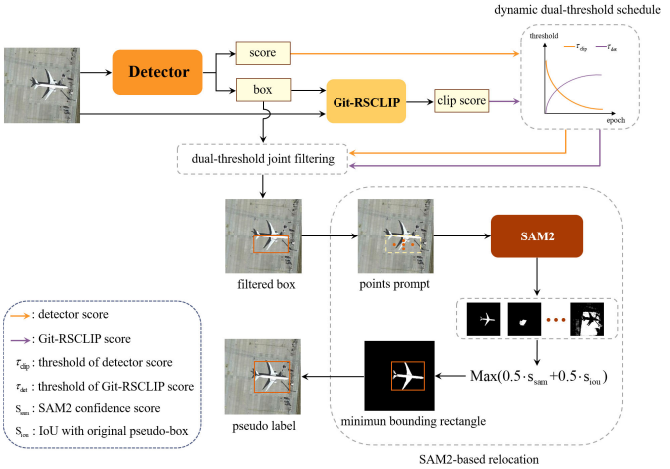


Fig. 4. Illustration of QPLR.

dual-threshold joint filtering strategy is adopted. Only when the bounding box simultaneously meets

$$s_{det} \geq \tau_{det}, s_{clip} \geq \tau_{clip} \tag{8}$$

will the resulting pseudo-label be retained, where τ_{det} and τ_{clip} are the filter thresholds for detector confidence and Git-RSCLIP score, respectively.

The detector concerned may achieve different performances at different training stages. Inspired by this observation, a dynamic dual-threshold scheduling strategy is further proposed to adapt to the model's learning process, such that

$$\tau_{det}(t) = \tau_{det}^{max} + \left(\tau_{det}^{max} - \tau_{det}^{min}\right) \cdot \exp(-\lambda t) \tag{9}$$

$$\tau_{clip}(t) = \tau_{clip}^{max} + \left(\tau_{clip}^{max} - \tau_{clip}^{min}\right) \cdot \exp(-\mu t) \tag{10}$$

where τ_{det}^{min} and τ_{det}^{max} represent the minimum and maximum values of τ_{det}, τ_{clip}^{min} and τ_{clip}^{min} represent the minimum and maximum values of τ_{clip}, λ and μ are smooth scheduling factors, and t represents the training step (epoch or iteration).

In the early stages of training, since the detector has not yet developed strong localization capabilities, a lower detector threshold τ_{det} is adopted with a higher Git-RSCLIP score threshold τ_{clip}. At such a stage, the filtering of pseudo-labels mainly relies on the semantic consistency evaluation of Git-RSCLIP to ensure the reliability of pseudo-label categories. As training progresses, the detector gradually gains more accurate object localization capabilities, and the detector threshold τ_{det} is gradually increased while the Git-RSCLIP score threshold τ_{clip} is gradually decreased. In the later stages of training, the localization quality of pseudo-labels is mainly controlled by the detector, while Git-RSCLIP plays more of an error-correction role, removing erroneous detection boxes that are clearly semantically inconsistent. This dynamic strategy effectively balances the semantic correctness and localization accuracy at different stages of training.

With dual-threshold joint filtering applied, retained pseudo-labels may still cause localization inaccuracies, including too much background area or covering only a partial region of the target. To address this problem, a relocation mechanism based

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on SAM2 [26] is also introduced to deal with pseudo-labels, in order to further improve their spatial accuracy.

For each pseudo-box, multiple independent point-prompted segmentations are performed inside it. In implementation, five point-prompted segmentations for each pseudo-box are executed, with the point locations at

$$P_c = (x_c, y_c) \tag{11}$$

$$P_{cl} = \left(x_c - \frac{w}{4}, y_c\right) \tag{12}$$

$$P_{cr} = \left(x_c + \frac{w}{4}, y_c\right) \tag{13}$$

$$P_{cb} = \left(x_c, y_c + \frac{h}{4}\right) \tag{14}$$

$$P_{cu} = \left(x_c, y_c - \frac{h}{4}\right) \tag{15}$$

where w and h represent the width and height of the pseudo-box, respectively. This design aims to increase the probability that the point-prompts fall within the actual target area, enhancing the stability of the segmentation results.

Since SAM2 generates three candidate segmentation results for each point-prompt segmentation, 15 segmentation masks can be obtained for each pseudo-box. Each segmentation mask is converted into its minimum bounding rectangle to yield 15 candidate relocation boxes. For each candidate box, both the segmentation confidence s_{sam} output by SAM2 and the intersection over union (IoU) value s_{iou} between the candidate box and the original pseudo-box are taken into account, with the final score being defined as

$$s = 0.5 \cdot s_{sam} + 0.5 \cdot s_{iou}. \tag{16}$$

The candidate box with the highest score is ultimately selected as the relocation result for the pseudo-box.

By introducing the dynamic dual-threshold joint filtering (DTJF) and the SAM2-based quality-aware relocation mechanism, the spatial localization accuracy of pseudo-labels is significantly improved while semantic consistency is ensured. This pseudo-label generation strategy continuously provides the model with high-quality object-level supervision during training, laying a solid foundation for subsequent detection performance improvement.

D. Image-Guided Query Enhancement

From a cognitive perspective, human visual recognition relies heavily on the direct perception of visual examples. Without encountering concrete instances of a particular category, relying solely on names or abstract descriptions often makes it difficult to form stable and accurate cognitive representations. However, once visual samples of that category are observed, their appearance, structural form, texture patterns, and morphological features can be quickly memorized and generalized, thereby significantly enhancing subsequent recognition ability. This phenomenon indicates that transmitting knowledge within the same visual modality space is generally more direct and efficient than cross-modal mapping.

Most existing OVOD methods work, however, by relying on language embeddings or text prompts to guide the detection process. Due to the inevitable modality gap between linguistic semantics and the visual feature representations within detection models, the corresponding guidance information tends to produce semantic shifts or insufficient expression when mapped into the detection feature space. This limits the model's ability to distinguish object instances. Inspired by this observation, an image feature-based query guidance mechanism is proposed herein. It directly introduces category-related visual priors to provide the detection model with guidance signals more consistent with its internal representation space, as shown in Fig. 5.

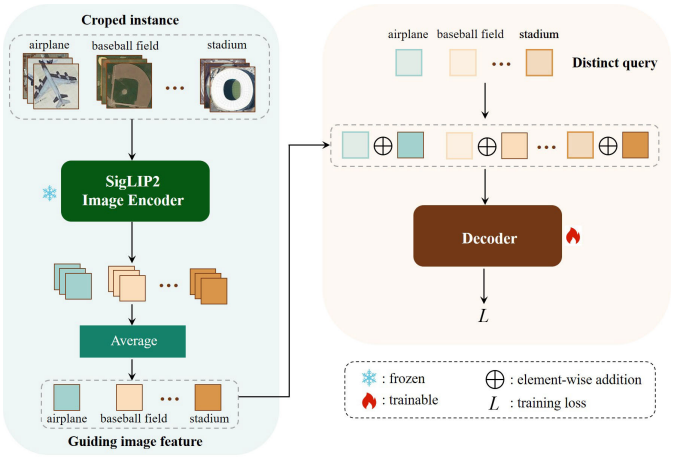


Fig. 5. Illustration of IGQE.

Particularly, the object-level pseudo labels obtained during training are first leveraged, and the SigLIP2 [49] image encoder is used to extract their object-level feature representations. For all instance features of the same category, feature mean aggregation is then adopted to construct the guided image feature. For category c, its guiding image feature is defined as

$$g_c = \frac{1}{N_c} \sum_{i=1}^{N_c} \phi\left(x_i^c\right) \tag{17}$$

where x_i^c represents the image region corresponding to the ith instance of category c, $\phi(\cdot)$ is the SigLIP2 image encoder, and N_c is the number of instances of category c. The image-guided features can effectively capture the common structural attributes and discriminative visual patterns of the category in terms of appearance, thereby providing stable and robust visual priors for subsequent detection.

In the original DDQ-DETR [48], the decoder's query vectors are generated by filtering and remapping the encoder features, and are used to progressively decode target instances. On this basis, category-level image-guided features are introduced to enhance the decoder query. Specifically, for each decoder query, the corresponding class-level image-guided features are used, exploiting its predicted category from the encoder. This constructs an enhanced query representation through linear fusion

$$\tilde{q}_i = \lambda \cdot q_i + (1 - \lambda) \cdot g_c \tag{18}$$

where q_i denotes the ith decoding query originally generated by DDQ-DETR, g_c is the image-guided feature for the

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category c in response to the query, and λ is a learnable scaling factor, initialized to 0.5 empirically. In this way, category-level image-guided features are injected into the decoder query in the form of linear fusion, explicitly enhancing visual patterns that are related to the target category without disrupting the original query structure.

The IGQE module as devised above uses category-level image-guided features to provide stable visual references for decoder queries, helping the overall detection model focus more accurately on target instances. This method fully leverages the expressive advantages of same-modality visual features, offering a more discriminative query modeling approach for OVOD in remote sensing scenarios.

IV. Experiments

A. Benchmark Datasets

A systematic evaluation of the proposed method is conducted on three commonly used remote sensing object detection benchmark datasets. To align with the experimental setup of OVOD, the categories in each dataset are divided into base classes and novel classes, training only on data annotated with base classes. During the validation phase, the detection performance is evaluated for both base and novel classes. The specific information and division methods of each dataset are as follows.

DIOR [53]: DIOR is a large-scale benchmark dataset for remote sensing image object detection, containing 23 464 images and 192 472 annotated instances. The image resolution is standardized to 800×800, covering a total of 20 categories. Under the OVOD setup, 15 categories are divided into base classes and five categories into novel classes. The novel classes include airport, airplane, chimney, overpass, and basketball court. The model is trained only on the training set with annotations of base class instances and tested on the validation set.

DOTA [54]: DOTA is a classic remote sensing object detection evaluation dataset, containing 2806 high-resolution remote sensing images and 188 282 annotated instances. The image sizes range from 800 to 4000 pixels in length, with a total of 15 categories. Ten of these categories are divided into base classes and five categories into novel classes. The novel classes are baseball diamond, roundabout, basketball court, storage tank, and plane. During the training phase, only the annotated data of the base classes is used, and the evaluation is conducted on both base and novel class objects in the validation set.

NWPU VHR-10 [55]: NWPU VHR-10 is a commonly used small-scale remote sensing object detection dataset, containing 800 images, of which 650 are annotated, with a total of 3896 instances. The image side length ranges from 358 to 1728 pixels and covers ten categories. Five categories are divided into base classes and five categories into novel classes. The novel classes include airplane, storage tank, baseball diamond, tennis court, and bridge. The dataset is split into training and validation sets at a 5:5 ratio. During training, only the training set with base class annotations is used, and testing is conducted on the validation set.

To introduce visual priors from image-level annotated data, the NWPU-RESISC45 [56] dataset is used as an auxiliary training data source. NWPU-RESISC45 is a large-scale remote sensing image classification dataset that provides only image-level annotations. It contains 45 semantic categories, covering a rich variety of remote sensing scene types, featuring high visual diversity and semantic discriminability. In this article, NWPU-RESISC45 is not used as an object detection evaluation dataset, but only as auxiliary data during the training phase. This is set in order to support image-level weak supervision for learning instance-level knowledge from image-level annotations.

B. Implementation Details

The proposed method uses DDQ-DETR [48] as the baseline model and extends it to handle OVOD by introducing SigLIP2 [49]. Specifically, the image encoder of SigLIP2 is adopted as the backbone network for DDQ-DETR, and the text encoder of SigLIP2 is used to extract the textual features corresponding to categories. Based on these textual features, a classifier is constructed to replace DDQ-DETR's original closed-set classification head, enabling the overall model to perform open-vocabulary recognition.

During the model training process, the AdamW [57] optimizer is used with a weight decay of 0.05 and an initial learning rate of $2e^{-4}$. The model is trained for 12 epochs on all datasets, with the learning rate decayed by a factor of 0.1 at the eighth and 11th epochs. The batch size during the training phase is uniformly set to 2. For data augmentation, only random horizontal flipping with a probability of 0.5 is applied. For the DOTA dataset, due to its high original image resolution, the common sliding window cropping strategy is adopted for data preprocessing. In particular, the original images in the training and validation sets are cropped into 800×800 image patches, with a 200-pixel overlap between adjacent patches to alleviate the problem of targets being possibly cut off. During the testing phase, inference is first performed on all image patches in the validation set, then the corresponding detection results are mapped and stitched back into the original image coordinate system, and finally, unified evaluation is conducted at the original scale. All experiments are implemented based on the PyTorch framework and conducted on NVIDIA GeForce RTX 5090 GPUs for training and evaluation.

During the evaluation phase, mean average precision (mAP) is systematically adopted with an IoU threshold of 0.5 as the primary performance metric. In all experiments, the mAP for all categories (mAP_a), for base classes (mAP_b), and for novel classes (mAP_n) are reported separately, to comprehensively evaluate the model's performance in open-vocabulary detection scenarios.

C. Experimental Results and Comparisons

1) On Dataset DIOR: The OVOD results on the DIOR dataset are shown in Table I. Observably, different methods exhibit significant performance differences on novel classes and base classes, reflecting their varying emphasis

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TABLE I
COMPARISON OF OVOD METHODS ON DATASET DIOR, WITH RED DENOTING BEST RESULTS AND BLUE REPRESENTING SECOND-BEST RESULTS

Methods	mAP_n	mAP_b	mAP_a
RegionCLIP [27]	18.64	72.27	58.86
PB-OVD [37]	16.92	72.09	58.30
VL-PLM [40]	36.56	82.68	71.15
WSOVOD [41]	40.62	87.50	75.78
CastDet [47]	39.68	88.93	76.61
Ours	55.18	90.07	81.35

TABLE II
COMPARISON OF OVOD METHODS ON DATASET DOTA, WITH RED DENOTING BEST RESULTS, AND BLUE REPRESENTING SECOND-BEST RESULTS

Methods	mAP_n	mAP_b	mAP_a
RegionCLIP [27]	11.28	56.31	41.30
PB-OVD [37]	7.44	56.13	39.00
VL-PLM [40]	11.26	59.17	43.20
WSOVOD [41]	12.80	59.24	43.76
CastDet [47]	11.50	60.21	43.90
Ours	15.14	67.43	50.00

TABLE III
COMPARISON OF OVOD METHODS ON DATASET NWPU VHR-10, WITH RED DENOTING BEST RESULTS AND BLUE REPRESENTING SECOND-BEST RESULTS

Methods	mAP_n	mAP_b	mAP_a
RegionCLIP [27]	42.80	41.60	42.20
PB-OVD [37]	32.58	36.48	34.50
VL-PLM [39]	41.98	46.22	44.10
WSOVOD [40]	47.88	67.60	57.74
CastDet [47]	45.08	65.74	55.40
Ours	49.36	82.64	66.00

on instance-level semantic modeling capability and cross-category generalization ability.

The mAP_n scores of novel classes achieved by RegionCLIP and PB-OVD are 18.64% and 16.92%, respectively, each showing limited performance. RegionCLIP enhances region-level semantic alignment through region–text pair fine-tuning, while PB-OVD introduces object-level supervision via a heatmap-based pseudo-label generation mechanism. However, both approaches largely rely on the direct semantic judgment of candidate regions by the VLM employed. This makes them difficult to consistently obtain high-quality instance localization results in complex remote sensing scenarios, constraining further performance improvements in novel classes detection.

VL-PLM and WSOVOD, and the recent remote sensing OVOD method CastDet, can achieve significant improvements in novel classes detection, with WSOVOD and CastDet attaining an mAP_n score of 40.62% and 39.68%, respectively. This indicates that object-level learning strategies based on pseudo-labels or weak supervision have advantages in alleviating the open-vocabulary transfer problem. Nonetheless, compared with the method proposed herein, these approaches still show a notable gap in overall performance and novel class generalization capability. This suggests that relying solely on a single form of weak supervision or pseudo-label mechanism remains insufficient to fully exploit the instance-level semantic information contained in image-level annotations.

Fortunately, the proposed method achieves superior performance on the DIOR dataset. For novel class detection, the mAP_n score reaches 55.18%, representing a substantial improvement over existing methods. Meanwhile, it maintains a high accuracy of 90.07% for base classes detection, resulting in an overall mAP_a increase to 81.35%. These results validate the effectiveness of the proposed modules—IWKI, QPLR, as well as IGQE—in effectively mining potential instance-level information from image-level annotations. Importantly, while greatly enhancing the generalization capability for novel categories, the proposed method does not sacrifice the detection performance of the base categories, achieving a more optimal balance between novel and base classes.

2) On Dataset DOTA: The results of OVOD on the DOTA dataset are shown in Table II. Compared with DIOR, the DOTA dataset features a larger range of resolution variations, more complex background structures, and a denser distribution of objects, imposing higher demands on OVOD methods in terms of instance-level semantic modeling and precise localization capabilities.

The mAP_n scores of RegionCLIP and PB-OVD for novel class detection are 11.28% and 7.44%, respectively, with the overall performance being relatively limited. This is mainly due to both methods relying heavily on the semantic similarity for candidate regions by the VLM used. On DOTA, where target scale varies drastically, orientation is pronounced, and background interference is severe, region discrimination based on global semantic similarity struggles to consistently capture instance-level spatial boundaries. This restricts improvements in novel class detection performance.

VL-PLM and WSOVOD, and CastDet achieve relatively better results in novel category detection, and, regarding the overall performance, with WSOVOD reaching 12.80% on mAP_n and CastDet reaching 43.97% on mAP_a. These results indicate that introducing object-level learning signals through pseudo-label or weak supervision mechanisms helps alleviate the generalization difficulties of open-vocabulary detection in complex remote sensing scenarios. However, performance remains limited because of insufficient depth in mining instance-level knowledge.

In comparison, the proposed method herein achieves the best overall performance on DOTA. For novel class detection, the mAP_n score reaches 15.14%, significantly outperforming the others. Meanwhile, it attains an mAP_b of 67.43% for base classes detection, increasing the overall mAP_a to 50.00%. These results verify the effectiveness of the proposed method in enhancing instance-level semantic modeling and spatial localization capabilities under complex remote sensing scenarios. Particularly on the highly challenging DOTA dataset, this method maintains a good balance between novel class generalization ability and base classes detection performance, demonstrating stronger robustness and generalization capability.

3) On Dataset NWPU VHR-10: The OVOD results on the NWPU VHR-10 dataset are shown in Table III. Compared

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with DIOR and DOTA, the NWPU VHR-10 has relatively fewer categories, but the differences in target scale are significant, and targets of the same category have strong consistency in appearance features and spatial distribution. Therefore, this dataset partially reduces the difficulty of semantic distinction between categories, but it still imposes clear requirements on instance-level localization accuracy and object boundary modeling capability.

The overall detection performance of RegionCLIP and PB-OVD on this dataset is relatively limited. The mAP_n and mAP_b scores of RegionCLIP are 42.80% and 41.60%, respectively, and the mAP_n and mAP_b of PB-OVD are 32.58% and 36.48%, respectively. The main reason for such limited performance is that both methods, when utilizing image-level supervision information, still heavily rely on the semantic matching of candidate regions by the VLM employed. This makes it difficult to fully characterize target instances in remote sensing images that have consistent structures and significant scale variations.

VL-PLM achieves relatively balanced performance in detecting novel and base classes, with its mAP_n and mAP_b scores reaching 41.98% and 46.22%, respectively. This suggests that the introduced pseudo-labeling mechanism helps enhance the model's instance perception ability. WSOVOD and CastDet exhibit competitive results, achieving an mAP_b of 67.60% and 65.74%, respectively, for base class detection and improving the overall performance to 57.74% and 55.40%. This demonstrates the effectiveness of weak supervision signals in improving remote sensing object detection performance.

In comparison, the proposed method achieves the best detection performance on the NWPU VHR-10 dataset. For novel class detection, the mAP_n reaches 49.36%, significantly outperforming the others; for base class detection, it achieves an mAP_b score of 82.64%, raising the mAP_a to 66.00%. These results fully validate that this method can effectively exploit the implicit object-level semantics and spatial information in image-level annotations, striking a more optimal balance between novel class generalization and base class detection performance, and demonstrating stronger robustness and generalization capability.

D. Ablation Study

To verify the contribution of each key module to the model's overall performance, systematic ablation experiments are conducted using DDQ-DETR as the baseline on the DIOR dataset, with the experimental results shown in Table IV. Without introducing any additional modules, the baseline model cannot detect novel categories. Although it maintains high accuracy in base class detection (mAP_b = 90.52%), the overall performance is significantly limited.

After introducing the IWKI module, the detection performance for novel classes is improved significantly, with the mAP_n rising to 38.72%. This result indicates that through global classification queries and with the DQP strategy, the model can effectively mine instance-level semantic information from image-level annotated data, providing strong support for OVOD.

TABLE IV
ABLATION STUDY RESULTS ON DATASET DIOR. BEST RESULTS ARE INDICATED IN BOLD FONT

IWKI	QPLR	IGQE	mAP_n	mAP_b	mAP_a
			-	90.52	67.89
✓			38.72	89.11	76.51
✓	✓		52.46	**90.07**	80.66
✓	✓	✓	**55.18**	**90.07**	**81.35**

On this basis, the QPLR module is further introduced, raising the novel classes detection performance to 52.46%, with the overall mAP_a correspondingly improved. This implies that high-quality pseudo-labels can provide the model with more reliable object-level supervision, while the dynamic dual-threshold filtering and the SAM2-based relocation mechanism effectively suppress the negative impact of noisy pseudo-labels over the training process.

When all three modules are enabled, that is, after introducing the IGQE module, the model achieves its best performance in novel class detection (mAP_n = 55.18%), resulting in an overall mAP_a score of 81.35%. This result indicates that the query-guided mechanism based on image features can further strengthen the detector's ability to model category-related visual patterns, complementing the previous two modules to achieve more robust and comprehensive performance improvements.

To intuitively demonstrate the specific roles and contributions of the proposed modules, a step-by-step visual ablation analysis is presented on novel categories as exhibited in Fig. 6.

1) *Limitation of the Baseline (Semantic Overfitting):* As shown in the second row, the baseline model completely lacks open-vocabulary recognition capabilities for novel classes. It exhibits severe semantic overfitting to base categories, forcibly misclassifying novel objects into morphologically similar base classes (e.g., airplane to windmill, airport to ground track field, and overpass to bridge).

2) *Role of IWKI (Semantic Breakthrough Versus Spatial Inaccuracy):* Upon integrating the IWKI module (see the third row), the model successfully acquires the ability to recognize novel categories. The semantic misclassifications are effectively corrected. However, since IWKI relies on image-level supervision to model object-level semantics without instance-level annotations, it struggles with precise spatial localization. Consequently, the predicted bounding boxes are either loose, fragmented, or encompass excessive background (e.g., partial coverage of the airport and background inclusion in the chimney).

3) *Role of QPLR (Spatial Rectification Versus Semantic Instability):* To address the spatial inaccuracy, the QPLR module is introduced (with the results given in the fourth row). By leveraging dual-threshold joint filtering and SAM2-based relocation, the localization precision is drastically improved. The bounding boxes for airplane, airport, and chimney now tightly encapsulate the objects. However, QPLR introduces slight semantic instability. For instance, an airplane is misclassified as a windmill because the initial pseudo-labels for novel classes may still be associated with a low confidence level. This causes the model to occasionally regress to

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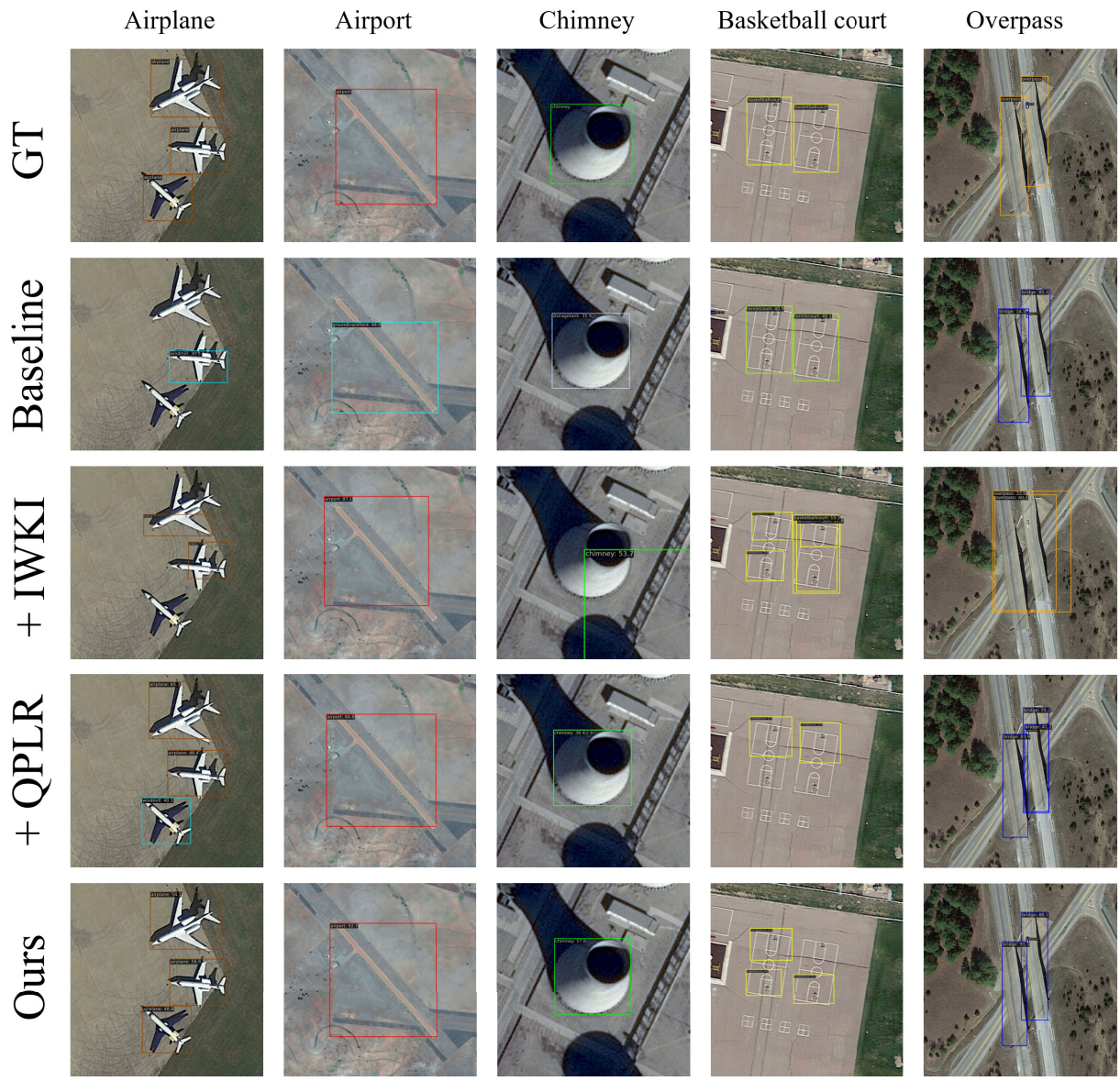


Fig. 6. Visual ablation analysis of detection results on the DIOR.

visually similar base classes during the spatial rectification process.

4) *Role of IGQE (Semantic Stabilization Confidence Boost):* The final model, incorporating the IGQE module (fifth row), resolves the aforementioned semantic instability. Using pre-extracted instance-level image features to guide the detector queries, IGQE provides stable and modality-consistent visual references. As observed, the misclassification of the airplane is now corrected, and the confidence scores across multiple categories (airport and chimney) are boosted, achieving high-quality detection that is both semantically accurate and spatially precise.

5) *Discussion on Limitations and Failure Cases:* Despite the overall significant improvements, the method still encounters challenges in certain extreme scenarios, as revealed in the last two columns as follows.

1) *Incomplete Localization:* The bounding boxes for the basketball court remain partially localized. This can be attributed to the inherent perception bias of Git-RSCLIP and the SAM2-based relocation mechanism. When prompted with points, SAM2 tends to segment regions with dense, closed boundary lines. Consequently, a half-court is often mistakenly perceived as an independent, complete instance.

2) *Persistent Misclassification:* The model still misclassifies the novel category overpass as the base category bridge. This is primarily due to the extreme morphological similarity between the two. Furthermore, the scarcity of high-quality pseudo-labels generated for overpass during the weakly supervised phase prevents the model from accumulating sufficient distinct instance-level knowledge, leading to an inevitable bias toward the dominant base category.

In summary, the results of the ablation experiments fully validate the effectiveness of each innovative module developed in this work for enhancing OVOD performance. These results empirically support the overall design rationality of the

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TABLE V

EFFICIENCY ANALYSIS OF QPLR. "DTJF" STANDS FOR DUAL-THRESHOLD JOINT FILTERING STRATEGY, "DDTS" FOR DYNAMIC DUAL-THRESHOLD SCHEDULING STRATEGY, AND "SBRM" FOR SAM2-BASED RELOCATION MECHANISM. BEST RESULTS ARE INDICATED IN BOLD FONT

DTJF	DDTS	SBRM	mAP_n	mAP_b	mAP_a
			38.72	89.11	76.51
✓			48.32	89.41	79.13
✓	✓		50.14	**90.13**	80.13
✓	✓	✓	**52.46**	90.07	**80.67**

proposed method and the complementary advantages among its modules.

E. Impact of Pseudo-Label Generation and Quality-Aware Relocation

To further verify the effectiveness of each key design within the QPLR module, additional step-by-step ablation experiments are conducted on the DIOR dataset, with the results summarized in Table V.

First, without introducing any pseudo-label filtering or relocation strategies, the model is trained solely with image-level knowledge injection. At this point, the mAP_n score for the novel category is 38.72%, and the overall detection performance is still significantly affected by noisy pseudo-labels.

Second, after introducing the dual-threshold joint filtering strategy, the mAP_n increases to 48.32%, with mAP_a also being raised from 76.51% to 79.13%. This indicates that the dual-threshold joint filtering strategy can effectively suppress low-confidence and semantically inconsistent pseudo-labels, reinforcing the overall quality of the pseudo-labels.

On this basis, by further adopting the dynamic dual-threshold scheduling strategy, the model's performance on novel categories is significantly improved to 50.14%, with both mAP_b and mAP_a increased accordingly. This result suggests that adaptively adjusting the thresholds as the training progresses can help maintain high-precision supervision in the early stages of training while introducing more high-quality pseudo-labels in the later stages, thereby achieving more thorough knowledge mining.

Finally, after introducing the SAM2-based relocation mechanism for refined spatial correction of pseudo-boxes, mAP_n further increases to 52.46%, and the overall mAP_a reaches 80.67%. These results verify the effectiveness of the quality-aware relocation strategy in both mitigating coarse box localization errors and strengthening target spatial alignment accuracy.

In summary, the above extensive experimental results reveal the fact that each component module introduced in this research, from pseudo-label selection and dynamic quality control to refined relocation, plays a complementary and indispensable role in performance improvement. The QPLR module can significantly enhance the reliability and spatial accuracy of pseudo-labels under weak supervision, providing high-quality supervisory signals for the stable optimization of subsequent detection models.

TABLE VI

COMPARATIVE EVALUATION OF COMPUTATIONAL EFFICIENCY AND DETECTION PERFORMANCE. THE EVALUATION IS CONDUCTED ON A SINGLE NVIDIA RTX 5090 GPU

Method	Training Time (Hours)	Inference Speed (FPS)	mAP_n	mAP_b	mAP_a
Baseline	8.93	17.5	-	90.52	67.89
LILK	17.23	17.5	55.18	90.07	81.35

F. Computational Efficiency and Tradeoff Analysis

To evaluate the practicality of the proposed approach, computational efficiency (in terms of both training time and inference speed) and detection performance are examined by comparing outcomes attainable with the baseline and the LILK framework. The results are summarized in Table VI. As can be seen, introducing the QPLR module inevitably increases the training time from 8.93 to 17.23 h. This additional computational overhead primarily stems from semantic consistency evaluation of Git-RSCLIP and SAM2-based relocation. However, this extra investment in time is a worthwhile tradeoff for two reasons as follows.

1) *Zero Overhead During Deployment:* The foundation models (SAM2 and Git-RSCLIP) are strictly utilized only during the pseudo-label generation and training phases. Once the model is trained, these auxiliary modules are entirely discarded. Consequently, the inference speed of LILK remains identical to that of the baseline (at 17.5 FPS), ensuring that real-time deployment efficiency is not compromised in practical remote sensing applications.

2) *Significant Capability Breakthrough:* In exchange for an additional 8.3 h of GPU training time, the proposed method achieves a fundamental breakthrough in OVOD. While the baseline completely fails to detect the novel categories of concern, LILK achieves an impressive mAP_n of 55.18%, substantially raising the overall mAP_a from 67.89% to 81.35% and maintaining competitive performance on the base categories (mAP_b). Considering the prohibitive financial and labor costs required to manually annotate instance-level bounding boxes for emerging novel categories, trading a few hours of machine computation for a fully automated, high-precision OVOD capability is highly cost-effective and practical.

G. Analysis of Dynamic Dual-Threshold Scheduling

To intuitively illustrate and empirically validate the proposed DTJF mechanism introduced in Section III-C, a systematic experimental analysis is conducted in terms of threshold variation curve, quantitative ablation, and qualitative visualization.

Threshold Variation Curve: As shown in Fig. 7, the trajectories of the dual thresholds over the training epochs are plotted. Following an exponential scheduling strategy, τ_{clip} drops rapidly in the early stages to transfer high-confidence semantic knowledge from Git-RSCLIP, while τ_{det} increases progressively as the detector matures. This intersection demonstrates the dynamic "knowledge handoff" process, replacing rigid empirical thresholds with a self-adaptive curriculum.

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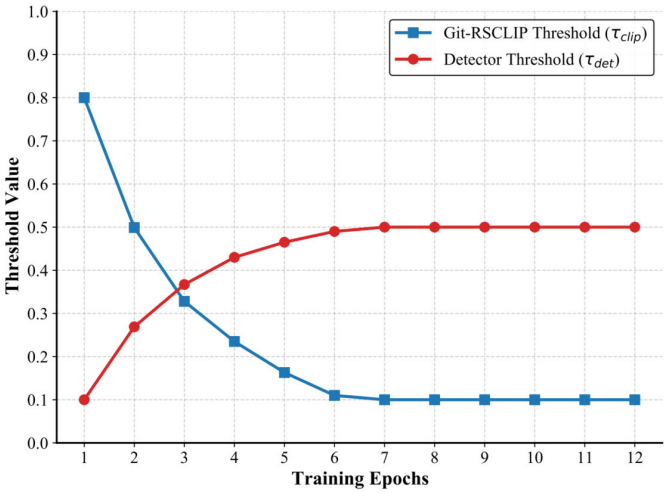


Fig. 7. Variation curves of dynamic dual-threshold scheduling over training epochs.

TABLE VII

ABLATION STUDY ON VARYING PSEUDO-LABEL FILTERING STRATEGIES, WHERE "DETECTOR" AND "GIT-RSCLIP" DENOTE RELYING ENTIRELY ON A SINGLE CONFIDENCE SOURCE, WITH BEST RESULTS INDICATED IN BOLD FONT

Detector	Git-RSCLIP	mAP_n	mAP_b	mAP_a
✓		42.35	**89.62**	77.80
	✓	44.76	89.07	77.99
✓	✓	**48.32**	89.41	**79.13**

Quantitative Ablation: To verify the effectiveness of the proposed joint scheduling, the DTJF module is compared with single-threshold baseline strategies on the DIOR dataset. As shown in Table VII, relying solely on the detector yields a limited novel class performance (mAP_n of 42.35%), as its initial ability to recognize unseen categories is rather limited. Using the Git-RSCLIP threshold solely improves novel class detection to 44.76%, but it remains struggling to model fine-grained spatial boundaries. By combining both through dynamic scheduling, DTJF establishes a mutually complementary mechanism, boosting the mAP_n to 48.32% while maintaining robust performance on base classes, achieving the best overall mAP_a of 79.13%.

Qualitative Visualization of Pseudo-Labels: Furthermore, the pseudo-labels generated under different strategies are visualized to qualitatively assess their quality. As illustrated in Fig. 8, relying entirely on the detector often leads to severe miss-detections of novel categories or overly loose bounding boxes. Conversely, single-filtering with Git-RSCLIP introduces bounding boxes that can successfully capture the semantic category but may lack spatial tightness (e.g., misaligned boxes or incorporating background interference). Fortunately, benefiting from DTJF, DTJF effectively filters out noise while preserving spatially precise and semantically accurate bounding boxes, providing higher quality pseudo-labels for training.

H. Analysis of Different VLMs

To empirically inform and justify the selection of the VLM in LILK, comparative experiments are conducted to evaluate the use of various mainstream VLMs as the visual and text encoders on the DIOR dataset.

TABLE VIII

COMPARISON OF USING DIFFERENT VLMS WHERE BEST RESULTS ARE INDICATED IN BOLD FONT

Architecture	Visual Encoder	Input Size	mAP_n	mAP_b	mAP_a
RemoteCLIP	ViT-Base	224×224	35.82	55.79	50.80
GeoRSCLIP	ViT-Base	224×224	33.46	53.61	48.57
SkyCLIP	ViT-Base	224×224	29.10	55.67	49.03
CLIP	ViT-Base	224×224	31.46	55.83	49.74
RemoteCLIP	ViT-Large	224×224	50.22	81.12	73.40
CLIP	ViT-Large	224×224	48.16	81.72	73.33
CLIP	ViT-Large-336	336×336	52.80	85.30	77.18
SigLip2	So400m	512×512	**55.18**	**90.07**	**81.35**

As illustrated in Table VIII, generic models (CLIP [21] and SigLIP2 [49]) and remote sensing-specific models (RemoteCLIP [32], GeoRSCLIP [23], and SkyCLIP [33]) are compared across different backbone sizes and input resolutions. While remote sensing-specific VLMs like RemoteCLIP show marginal advantages over standard CLIP at the base resolution of 224 × 224, the overall detection performance for all models investigated at this resolution remains limited (hovering around 50% for mAP_a).

The fundamental cause of this limited performance is that OVOD in remote sensing imagery is exceptionally sensitive to input resolution. Remote sensing targets are typically characterized by small scales, dense distributions, and arbitrary orientations. Formatting images to a low resolution of 224 × 224 aggressively compresses the underlying visual information, forcing small instances to collapse into background pixels and therefore destroying the fine-grained spatial details required for accurate bounding box localization. This is evident with the results of scaling the input size: by simply increasing the input resolution of the CLIP (ViT-Large) model from 224×224 to 336×336, the performance on novel classes (mAP_n) leaps from 48.16% to 52.80%.

Consequently, SigLIP2 emerges as the most suitable foundation model for LILK. Its So400m architecture supports a massive high-resolution input of 512 × 512. By maximally preserving the spatial granularity and critical edge features of remote sensing imagery, SigLIP2 resolves the problem of semantic-spatial misalignment prevalent in low-resolution models. Coupled with our proposed methodology, this architecture strikes an optimal balance, achieving the state-of-the-art performance with an mAP_n of 55.18% and an overall mAP_a of 81.35%.

I. Model Complexity Analysis

To evaluate the practical deployment feasibility of the proposed method (LILK), its computational complexity and parameter scale are compared with popular remote sensing OVOD baselines over the inference phase on the DIOR dataset. The number of parameters (Params) and giga floating-point operations per second (GFLOPs) are reported, as detailed in Table IX. As can be seen from this table, different

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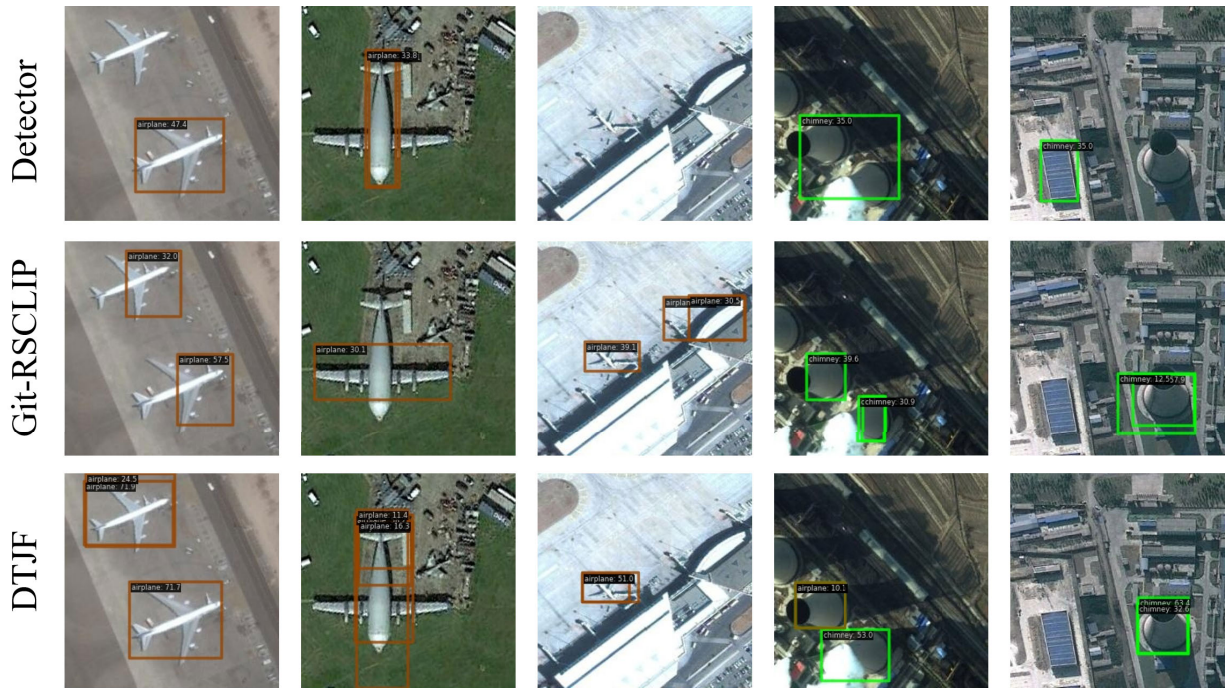


Fig. 8. Qualitative visualization of generated pseudo-labels across different filtering strategies on NWPU-RESISC45, as compared to relying exclusively on the detector alone or Git-RSCLIP; DTJF excels in generating more comprehensive and spatially precise bounding boxes.

TABLE IX
COMPARISON OF MODEL COMPLEXITY (PARAMS AND GFLOPS) AND DETECTION PERFORMANCE DURING INFERENCE PHASE, WHERE BEST RESULTS ARE INDICATED IN BOLD FONT

Methods	Params(M)↓	GFLOPs↓	mAP$_n$↑	mAP$_b$↑	mAP$_a$↑
WSOVOD	430	1235	40.62	87.50	75.78
CastDet	**161**	**183**	39.68	88.93	76.61
LILK (Ours)	456	466	**58.18**	**90.07**	**81.35**

methods adopt distinct strategies that significantly impact their inference complexity. WSOVOD achieves baseline open-vocabulary capabilities by directly embedding SAM into its architecture. However, relying on SAM during the online inference phase leads to an enormous computational burden, reaching 1235 GFLOPs, which heavily restricts its real-time deployment potential.

Conversely, CastDet utilizes a dual-teacher mechanism exclusively during model training. By discarding the teachers over the inference process, it maintains a highly lightweight profile (161 M Params, 183 GFLOPs). However, this lightweight design comes at the cost of limited generalization, resulting in a suboptimal novel class performance (mAP$_n$ of 39.68%).

In contrast, the proposed method (LILK) strikes an optimal balance. Note that the heavy multimodal modules (i.e., Git-RSCLIP and SAM2) utilized for pseudo-label optimization in this framework are operated entirely offline during the training phase and are completely decoupled from inference. The Params and GFLOPs reported in Table IX for the proposed method primarily originate from utilizing SigLIP2 as the powerful visual encoder for the online detector.

While this introduces a moderate increase in computation compared to CastDet (466 versus 183 GFLOPs), it is approximately 60% more efficient than WSOVOD. This acceptable investment in the SigLIP2 backbone translates into a massive, unprecedented absolute gain of + 15.5% in novel class mAP over CastDet. Therefore, LILK achieves a state-of-the-art tradeoff between deployment efficiency and open-vocabulary detection capability.

J. Visualization Analysis

To help evaluate the practical performance of the proposed method in OVOD for remote sensing, a visual analysis of detection results is further conducted on the DIOR dataset. The experimental results are illustrated in Fig. 9, where the first row represents ground truth, the second shows the detection results of WSOVOD [40], and the third presents those of the proposed method, LILK. From left to right, the figure displays five novel categories: airport, airplane, chimney, basketball court, and overpass.

In the airport category, both WSOVOD and LILK can correctly identify and accurately locate the target. However, compared to WSOVOD, LILK exhibits more stable responses in the target area with a higher prediction confidence value, indicating stronger discriminative capability in modeling the overall structure of large-scale targets.

In the airplane scene, LILK can comprehensively and accurately detect multiple target instances, whereas the use of WSOVOD clearly exhibits both false negative and false positive detections. This indicates that the proposed method can attain stable instance-level detection performance in multi-instance remote sensing scenarios.

For the chimney category, LILK significantly outperforms WSOVOD in terms of target localization accuracy, with its predicted boxes aligning more closely to the true boundaries

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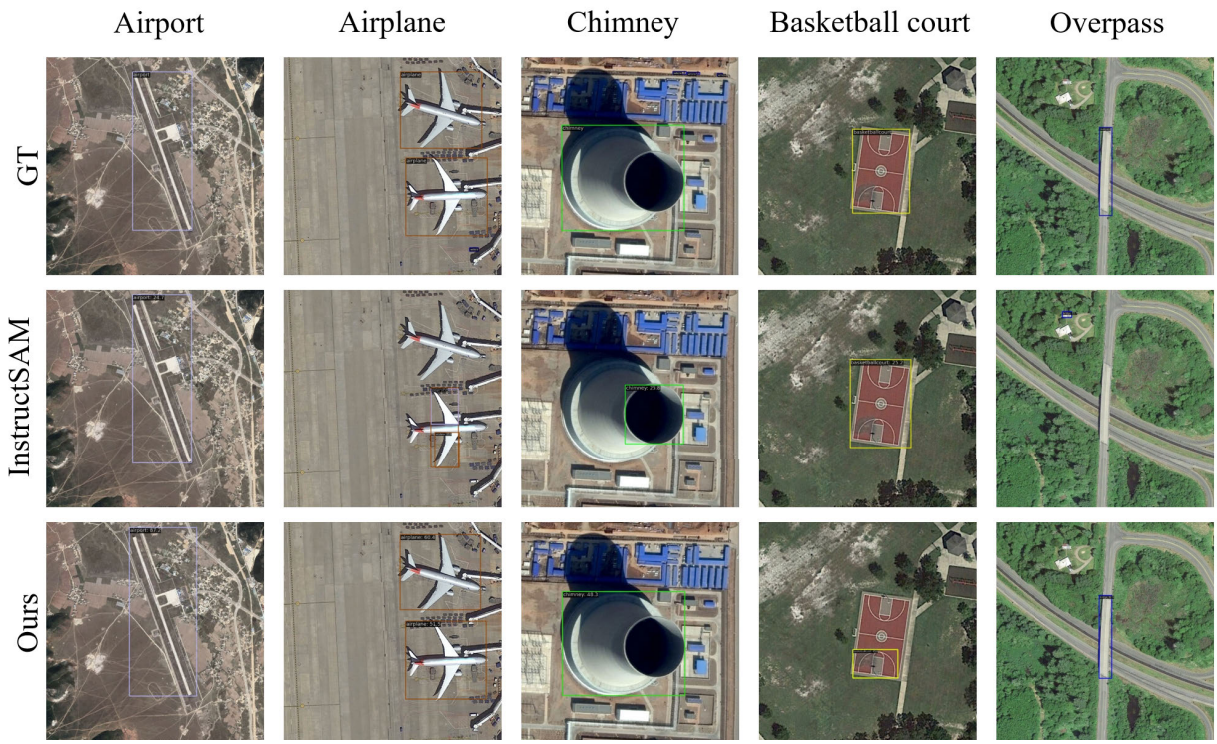


Fig. 9. Visualization of detection results on dataset DIOR.

of the target. This validates the effectiveness of the proposed QPLR module in improving spatial localization accuracy.

In the basketball court scenario, WSOVOD's detection results are relatively more complete, whereas LILK can only cover part of the target area. This phenomenon mainly stems from the limitations of Git-RSCLIP [24] and SAM2 [26] in modeling the basketball court category in remote sensing scenes: on one hand, Git-RSCLIP tends to assign a higher confidence value to locally discriminative regions inside the basketball court rather than modeling the full structure of the venue; on the other hand, under point prompt conditions, SAM2 struggles to accurately recover the overall area of a basketball court (a target with regular geometric boundaries but relatively complex semantic textures). Even when the prompt point is located inside the target, it may only segment a partial region.

In the overpass category, both methods face certain challenges. WSOVOD suffers from missed (or false negative) detections of foreground objects and misidentification of background areas as targets (i.e., false positives), while LILK tends to misclassify overpasses as bridges. This is mainly because the two types of objects have highly similar structural features in remote sensing images. During training, the bridge category has a larger number of samples with higher annotation quality, which leads the model to develop a certain bias toward bridges in visual-semantic discrimination.

Overall, the above visualization results indicate that LILK can achieve more stable and accurate instance-level detection performance in most novel category scenarios. At the same time, the existing failure cases, which reveal the method's limitations in certain categories, particularly in the fine-grained distinction between classes with highly similar appearances, as well as in the complete region modeling of targets with complex internal structures. These analyses provide valuable references for further optimizing pseudo-label quality modeling and query-guided strategies in the future.

K. Robustness Analysis of Pseudo-Label Generation in Challenging Scenarios

To evaluate the reliability of the pseudo-label generation pipeline, a qualitative analysis of the pseudo-label quality is carried out under three inherently challenging scenarios in remote sensing, respectively, involving noisy images, small targets, and dense targets, as illustrated in Fig. 10. These scenarios remain open challenges in the broader field of object detection. The detector is trained on the base class data of DIOR and also on the auxiliary dataset (NWPU-RESISC45).

1) *Noisy Images:* As shown in the first two rows of Fig. 10, severe environmental noise inevitably degrades the feature representation of pretrained models. Consequently, ambiguous backgrounds are occasionally misidentified as objects (identified with red ellipses), and obscured targets fail to be activated (signified by blue ellipses). However, despite the interference, LILK successfully extracts accurate bounding boxes for most distinct objects. This demonstrates that the QPLR module effectively prevents a complete semantic collapse, exhibiting reasonable resilience against complex meteorological and sensor noise.

2) *Small Targets:* The middle two rows demonstrate the performance on small targets. This is the most challenging scenario for the pseudo-label generation pipeline. Due to limited pixel areas, the foundation models (Git-RSCLIP and

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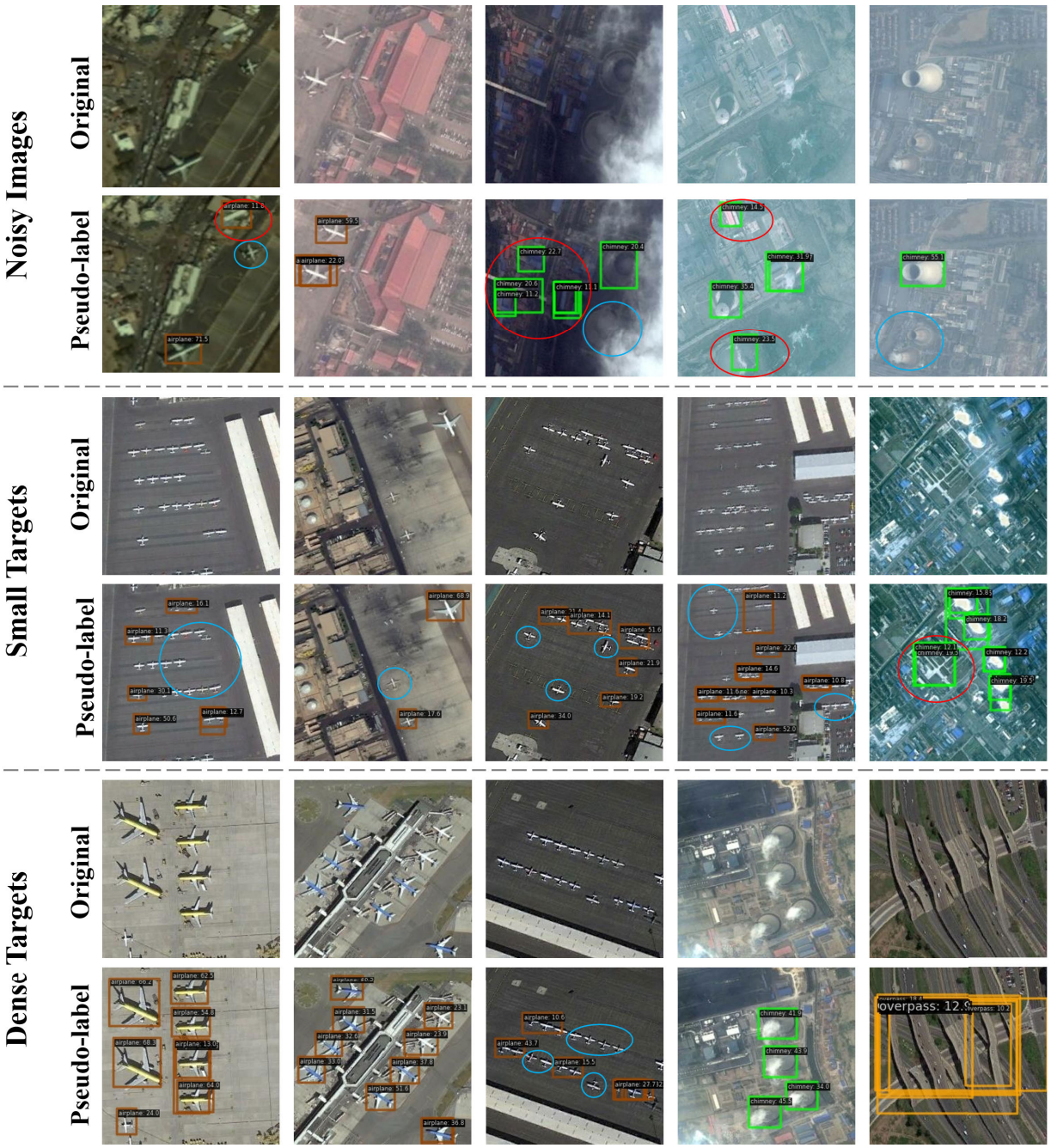


Fig. 10. Visual analysis of pseudo-label generation under three challenging scenarios: noisy images, small targets, and dense targets with images sampled from NWPU-RESISC45, where red ellipses indicate false positives, while blue ellipses indicate false negatives.

SAM2) struggle to capture sufficient discriminative semantics and spatial boundaries, leading to a notably high false-negative rate (frequent blue ellipses). The miss rate correlates directly with the reduction in target size. Nevertheless, from the perspective of weakly supervised learning, low recall is acceptable provided that precision is maintained. As observed, the targets that are successfully detected retain certain accurate bounding boxes, ensuring that the detector learns reliable, albeit sparse, instance-level knowledge without being misled by hallucinated labels.

3) *Dense Targets:* The last two rows illustrate the performance on dense targets. For medium-to-large dense objects (e.g., airplane in columns 1 and 2, and chimney in column 4), LILK maintains a high accuracy. However, performance drops significantly when dealing with "dense + small" targets (column 3), resulting in missing detections. Furthermore, for categories with lower structural distinctiveness (e.g., overpass in column 5), Git-RSCLIP suffers from cognitive limitation, grouping multiple closely packed instances into a single large bounding box.

Overall, while the inherent limitations of foundation models lead to a noticeable drop in pseudo-label quality under all three extreme scenarios, the proposed LILK framework does not catastrophically fail. The QPLR module acts as a conservative

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safeguard ensuring a certain amount of accurate detection boxes in the generated pseudo-labels.

V. Conclusion

This article has addressed the issues of insufficient instance-level localization and semantic alignment capabilities in OVOD in remote sensing images. It has proposed an OVOD framework, named LILK, that learns instance-level knowledge from image-level data. It effectively mines the implicit instance-level semantic information in image-level annotations through weak supervision. In so doing, LILK helps deal with challenging problems faced by existing methods, including heavy reliance on visual-language models for instance-level localization while being constrained by the underlying global semantic modeling paradigm, while makes precise spatial alignment difficult, LILK effectively mines the implicit instance-level semantic information in image-level annotations through weak supervision. This significantly enhances the localization accuracy and open-vocabulary generalization capability of object detection without introducing additional instance-level annotations.

Extensive experimental investigations have been carried out on three representative remote sensing object detection datasets, DIOR, DOTA, and NWPU VHR-10. The results of comparative studies systematically demonstrate that LILK achieves advanced performance in base classes, novel classes, and overall detection performance, validating its effectiveness and feasibility. The visualization analysis results performed also indicate that whilst LILK can achieve significant improvements in overall performance, there remain detection deficiencies in certain object categories. Future research will focus on these categories, exploring more effective instance-level feature modeling and optimization strategies to further enhance the model's detection performance.

Finally, while LILK has demonstrated strong open-vocabulary capabilities, it can be observed that categories with extremely high visual similarity (e.g., overpasses and bridges) remain prone to misclassification. As an important piece of future research, text-based prompt engineering can be exploited to inject fine-grained contextual knowledge into the vision-language alignment, with an aim to further enhance the discriminative ability when handling such easily confused categories.

Conflict of Interest Statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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